

Notes: Rotemberg and Woodford (JPE, 1992)

Oligopolistic Pricing and the Effects of Aggregate Demand on Economic Activity

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1 Big picture

- **Question.** How can aggregate demand shocks, especially government military purchases, raise output, hours, and real wages in a general-equilibrium model?
- **Puzzle.** In a perfectly competitive model, a demand shock cannot shift firms' labor demand at a given real wage. It can raise employment only through labor supply. But the postwar military-purchase evidence suggests that private output and real wages rise together.
- **Core answer.** Product-market power matters. If firms set price above marginal cost, the labor-demand condition becomes markup-adjusted. A decline in the markup shifts labor demand outward, allowing output, employment, and real wages to rise together.
- **Mechanism.** Industries are oligopolies sustained by implicit collusion. A current demand boom raises the temptation to undercut rivals today relative to future punishment value. To preserve incentive compatibility, the optimal collusive agreement lowers the markup in high-demand states.
- **Quantitative exercise.** The paper embeds this Rotemberg-Saloner-style oligopoly in a dynamic general-equilibrium model, calibrates it, linearizes it around steady state, and compares its impulse responses with those from a similarly calibrated competitive model and with empirical impulse responses to military spending shocks.
- **Main empirical comparison.** Both the competitive and oligopolistic models can match the private-output response reasonably well. The key difference is real wages: the competitive model predicts a decline, while the oligopolistic model predicts an increase that lies within the empirical confidence band.
- **Takeaway.** Countercyclical markups generated by imperfect competition provide a mechanism through which aggregate demand shocks have real effects that look Keynesian while remaining embedded in intertemporal general equilibrium.

2 Incremental contribution of the paper

- **Dynamic oligopoly inside general equilibrium.** The paper embeds a repeated-game model of oligopolistic collusion into an RBC-style dynamic general-equilibrium environment, rather than treating markups as exogenous wedges.
- **Microfoundation for countercyclical markups.** Markups fall in booms because current demand raises the payoff from undercutting collusive prices. The collusive price must fall to satisfy firms' incentive constraints.
- **Demand shocks can shift labor demand.** The paper shows that aggregate demand shocks can move the labor-demand curve through endogenous markup variation, not only through labor supply or technology.
- **Empirical discipline from military spending.** The authors use innovations in military purchases as relatively exogenous aggregate demand shocks and compare model impulse responses to empirical impulse responses.
- **Competitive benchmark comparison.** The contribution is not just showing that the oligopoly model can fit some facts; it directly compares the calibrated oligopolistic economy with a calibrated competitive economy using the same preference and technology environment.
- **Real wage evidence.** The paper emphasizes a fact that is difficult for standard competitive models: real wages rise after military demand shocks. The oligopoly model explains this through falling markups.
- **Calibrated markup conservatism.** The paper intentionally uses a conservative steady-state markup of about 1.2, showing that even modest market power improves the model's fit.
- **Connection to New Keynesian macro.** It gives a real, strategic-pricing route to demand-driven fluctuations, complementing nominal rigidity explanations of aggregate demand effects.

- **Historical robustness.** The paper checks whether the output/wage response also appears in a longer sample including major wars, adding credibility to the postwar evidence.

3 Data and measurement

3.1 Military-purchase shocks

- The empirical demand shock is an innovation in military purchases, chosen because military spending is plausibly more exogenous than many other components of government spending.
- The postwar empirical work separates military purchases and military employment, then maps them into the model's government-purchases and government-hours variables.
- The key shock is the component of military purchases that raises government demand for goods and services.
- A second orthogonal shock changes the composition of military spending toward military employment; the paper treats this component more cautiously.

Interpretation: The empirical design tries to isolate shocks to product demand, not shocks to technology or private labor supply. Military purchases are not perfect instruments, but they are a useful aggregate demand series because many movements are driven by geopolitical events rather than current macroeconomic conditions.

3.2 Main empirical outcomes

- **Private value added:** the private-sector output response to military demand shocks.
- **Private hours:** labor input in the private sector.
- **Real wages:** real hourly earnings, particularly manufacturing earnings, deflated by relevant price deflators.
- **Historical robustness variables:** annual changes in real defense purchases, real GNP, and real wages from a longer historical sample.

Interpretation: The key empirical target is joint behavior. A model must not only make output rise; it must also match the relative movements of hours and real wages.

3.3 Calibration objects

- Table 1 collects calibrated steady-state shares, elasticities, the steady-state markup, labor-supply parameters, and the parameter governing markup sensitivity.
- Some values are standard macro calibration inputs, such as the investment share, capital share, depreciation rate, interest rate, and capital-output ratio.
- Other values discipline oligopoly behavior, including the steady-state markup and elasticity of markup with respect to demand.

Interpretation: The calibration deliberately avoids choosing parameters solely to match the military-shock responses. This makes the comparison between model and data more informative.

4 Models and theoretical specifications

4.1 Competitive benchmark

In a competitive economy, labor demand satisfies

$$F_H(K_t, H_t, z_t) = w_t,$$

where K_t is predetermined capital, H_t is labor, z_t is technology, and w_t is the real wage. Holding capital and technology fixed, product demand does not shift this labor-demand curve.

Interpretation: This is the central problem for the competitive model. If military spending raises employment, it must do so through labor supply. But a pure labor-supply explanation tends to imply falling real wages, contrary to the evidence.

4.2 Imperfect competition and markup-adjusted labor demand

With product-market power, the first-order condition becomes

$$F_H(K_t, H_t, z_t) = \mu_t w_t,$$

where μ_t is the price-to-marginal-cost markup. A fall in μ_t shifts labor demand outward.

Interpretation: The aggregate demand shock can now affect production even if capital and technology are fixed, because demand changes the markup chosen by firms.

4.3 Oligopolistic pricing and incentive compatibility

Firms within an industry implicitly collude. They maintain high markups because deviation is punished by future lower profits. In a high-current-demand state, the short-run gain from undercutting rivals rises. Therefore, the maximum sustainable markup falls.

High current demand $\Rightarrow$ stronger temptation to deviate $\Rightarrow$ lower incentive-compatible markup.

Interpretation: The mechanism is a dynamic strategic-pricing mechanism. The markup is not assumed to fall; it falls because the collusive agreement must remain self-enforcing.

4.4 Dynamic general equilibrium

The oligopoly block is embedded in a general-equilibrium model with consumption, investment, capital accumulation, government demand, and labor supply. The authors log-linearize equilibrium conditions around steady state and solve for impulse responses.

Interpretation: The dynamic structure matters because collusion is forward-looking. Firms compare current deviation gains with future punishment losses, and households respond intertemporally to government spending shocks.

4.5 Model comparison

The paper computes impulse responses for:

- a competitive model with standard price-taking firms;
- an oligopolistic collusion model with countercyclical markups;
- empirical postwar impulse responses with two-standard-error confidence bands.

Interpretation: The comparison isolates what is gained by adding oligopolistic pricing to an otherwise similar dynamic general-equilibrium structure.

5 Identification and empirical design

5.1 Why military spending is used

Military purchases are closer to exogenous product-demand shocks than many other components of government spending. They are often linked to geopolitical events and defense policy rather than contemporaneous private-sector output or labor-market conditions.

5.2 Impulse-response design

- The paper estimates how private output, private hours, and real wages respond over several periods following military spending shocks.
- The same military-spending process is fed into the calibrated competitive and oligopolistic models.
- The empirical and theoretical impulse responses are then compared graphically.

Interpretation: This is a model-validation design rather than a conventional microeconomic causal design. The paper asks which structural model better reproduces the empirical dynamic response of the macroeconomy.

5.3 Threats to interpretation

- Military spending can be associated with patriotism, labor-supply shifts, or wartime controls.
- The military-employment component is harder to interpret than the purchase component.
- Historical prewar data are lower quality than postwar data.
- The oligopoly model is calibrated rather than fully estimated.

Interpretation: The paper is strongest when focusing on the postwar purchase shock and the real-wage response. It is more cautious about composition shocks and prewar robustness.

6 Section-by-section study guide

- **Introduction.** States the central empirical fact: military spending increases private output and real wages, which competitive models have difficulty explaining.
- **Section I.** Explains why competitive labor demand cannot shift with aggregate demand and introduces variable markups as the key mechanism.

- **Section II.** Builds the oligopolistic pricing model with differentiated products, repeated interaction, and self-enforcing collusion.
- **Section III.** Embeds the pricing model in a dynamic general-equilibrium economy and derives log-linear equilibrium conditions.
- **Section IV.** Calibrates the model parameters, including macro shares, labor-supply parameters, and the steady-state markup.
- **Section V.** Estimates postwar responses to military-spending innovations.
- **Section VI.** Compares empirical impulse responses to competitive and oligopolistic simulations.
- **Section VII.** Checks longer historical evidence using wars and prewar annual data.
- **Section VIII.** Discusses alternative markup mechanisms and the implications for macroeconomic theory.

7 Tables and figures

7.1 Table 1: Calibrated parameters

Read:

- Table 1 reports the calibrated parameters used in the simulations.
- It includes growth, output-share variables, depreciation, interest rates, labor-supply elasticities, the steady-state markup, and the parameter governing collusive markup response.
- The steady-state markup is set to $\mu = 1.20$, which the authors describe as conservative.
- The elasticity of markup with respect to demand is set to 0.19, below the limit implied by incentive compatibility.

Detailed interpretation: The calibration table is central because all model predictions depend on these values. The key strategic-pricing parameters are deliberately modest, so the paper is not simply using extreme market power to fit the data. The macro shares tie the model to U.S. national-account magnitudes, while the oligopoly parameters determine how strongly markups respond to demand.

Economic message: Even conservative market power can substantially improve the model’s ability to match real-wage dynamics after demand shocks.

TABLE 1 CALIBRATED PARAMETERS			
Parameter	Defined by (1)	Values (2)	Description (3)
	$\gamma_c \gamma_N - 1$	.008	Steady-state growth rate (per quarter)
	C/Y	.697	Share of private consumption expenditure in private value added
	G/Y	.117	Share of government purchases of goods in private value added
	$(g + \delta)(K/Y)$	.186	Share of private investment expenditure in private value added
		.013	Rate of depreciation of capital stock (per quarter)
	$\hat{H}/\hat{H}$	.17	Share of total hours supplied to government sector
	$F_H \hat{H}/F$	.75	Share of labor income in private value added
	$F_K \hat{K}/F$	.25	Share of capital in private value added
	$(F_K/\mu) - \delta$ or $\gamma_c^* \beta^{-1} - 1$	.015	Steady-state real rate of return (per quarter)
	$F_K F_H / F_{KH} F$	1	Elasticity of substitution between capital and hours
		1, .33	Elasticity of consumption growth with respect to real return, with hours worked held constant
		6, 1.30	Intertemporal elasticity of labor supply
		1, 1.20	Steady-state markup (ratio of price to marginal cost)
	$\bar{X}_\mu / \bar{Y}_\mu$	0, .19	Elasticity of the markup with respect to X/Y
		.9	Probability that collusion will continue in next quarter

Table 1: Calibrated parameters.

7.2 Figures 1 and 2: Private output and private hours after a military spending shock

Read:

- Figure 1 compares the response of private value added in the data, the competitive model, and the collusive oligopoly model.
- Figure 2 does the same for private hours worked.
- Both models can match the positive private-output response fairly well.
- Hours rise less than output in the data, implying procyclical productivity.

Detailed interpretation: The first two figures show that output alone is not enough to distinguish the models. The competitive model can generate a positive output response under certain labor-supply assumptions. The difficulty is the joint movement of output, hours, productivity, and wages. The hours response is smaller than the output response, so the model must also account for rising measured productivity.

Economic message: Output and hours responses are necessary but not sufficient tests; the real wage response is the decisive evidence.

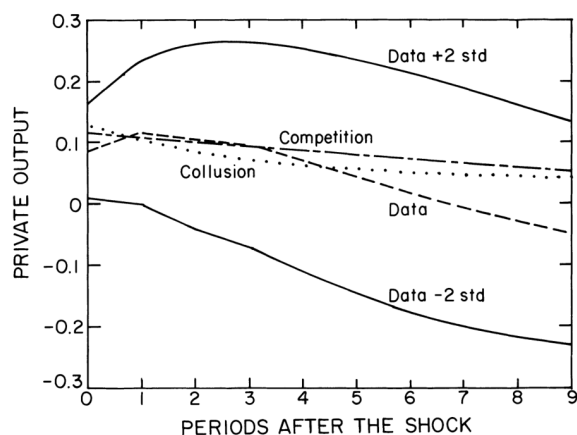


FIG. 1.—Response of private value added to a military spending shock

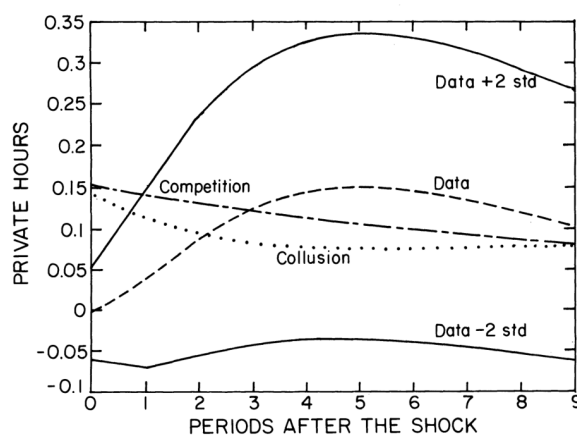


FIG. 2.—Response of private hours to a military spending shock

Figure 1: Private value added after a military-spending shock.

Figure 2: Private hours after a military-spending shock.

7.3 Figure 3: Real wages after a military spending shock

Read:

- Figure 3 compares real-wage responses in the data, competitive model, and collusive model.
- The empirical real wage rises after the military demand shock.
- The competitive model predicts a decline in real wages.
- The oligopolistic model predicts an increase and remains inside the empirical confidence band.

Detailed interpretation: This is the paper's most important figure. In the competitive model, employment increases require labor supply to shift outward, which tends to lower real wages. In the oligopolistic model, the falling markup shifts labor demand outward, so hours and real wages can rise together. The figure therefore distinguishes demand-through-labor-supply from demand-through-markups.

Economic message: Countercyclical markups allow Keynesian-looking demand effects without nominal rigidities.

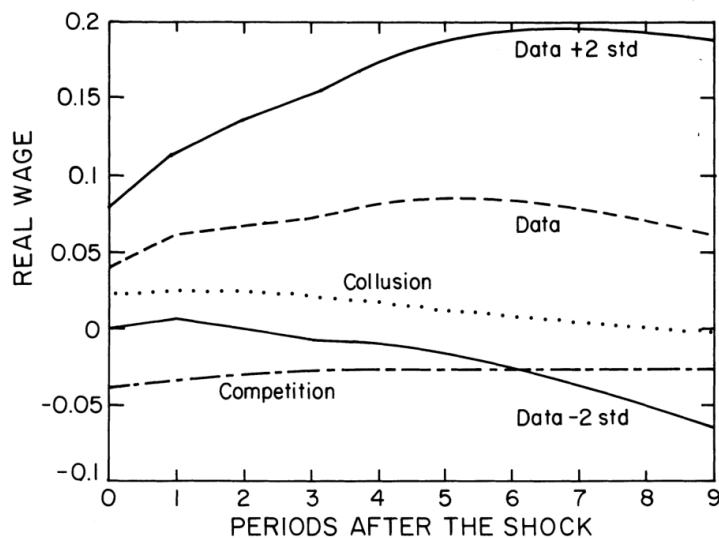


FIG. 3.—Response of real wages to a military spending shock

Figure 3: Real wages after a military-spending shock.

7.4 Figures 4 and 5: Sensitivity to markup level and markup elasticity

Read:

- Figure 4 shows how the first-period output response changes with the steady-state markup μ and the markup-response parameter ε_μ .
- Figure 5 shows the corresponding real-wage response.
- Output and real wages rise more when the steady-state markup is larger.
- Output and real wages also rise more when the markup is more sensitive to demand.

Detailed interpretation: These sensitivity surfaces show that the strategic-pricing mechanism works through two margins: the size of the wedge between price and marginal cost and the cyclical flexibility of that wedge. A higher markup means more room for a demand-driven decline in markups; a higher ε_μ means the incentive-compatibility constraint bites more strongly in booms.

Economic message: The model's success depends on countercyclical markups, not merely on intertemporal preferences.

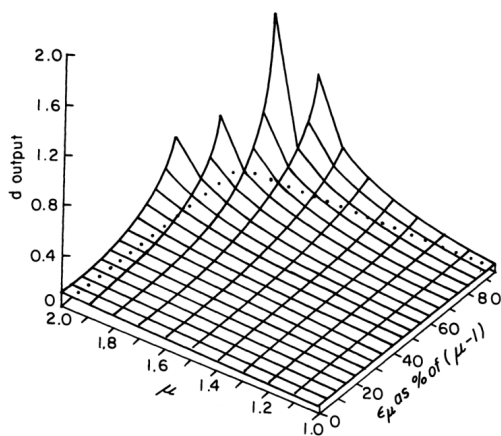


FIG. 4.—Sensitivity of the first-period output response to μ and ϵ_μ

Figure 4: Output sensitivity to μ and ϵ_μ .

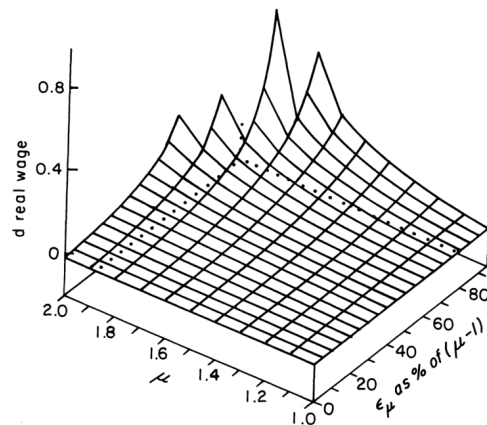


FIG. 5.—Sensitivity of the first-period real-wage response to μ and ϵ_μ

Figure 5: Real-wage sensitivity to μ and ϵ_μ .

7.5 Figures 6 and 7: Sensitivity to preference parameters

Read:

- Figure 6 shows how the first-period output response changes with intertemporal substitution and labor-supply elasticity parameters.
- Figure 7 shows the corresponding real-wage response.
- Large labor-supply responses can raise output in the competitive model.
- But large labor-supply responses tend to lower real wages, which conflicts with the empirical evidence.

Detailed interpretation: These figures show why preferences alone cannot solve the puzzle. If government spending raises output through labor supply, wages should fall. The data show rising real wages. This makes the model favor a labor-demand shift through lower markups rather than a pure labor-supply shift.

Economic message: The empirical real-wage response rules out explaining the military-spending multiplier only with labor-supply movements.

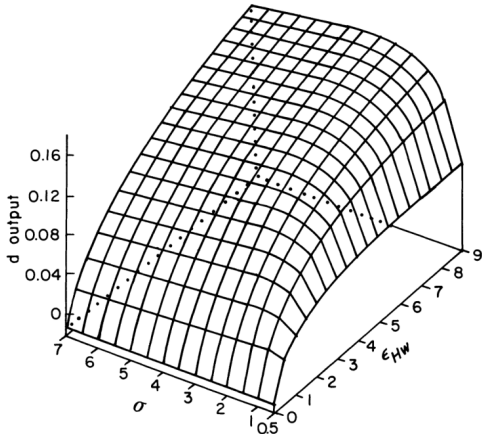


FIG. 6.—Sensitivity of the first-period output response to σ and ϵ_{Hw}

Figure 6: Output sensitivity to σ and ϵ_{Hw} .

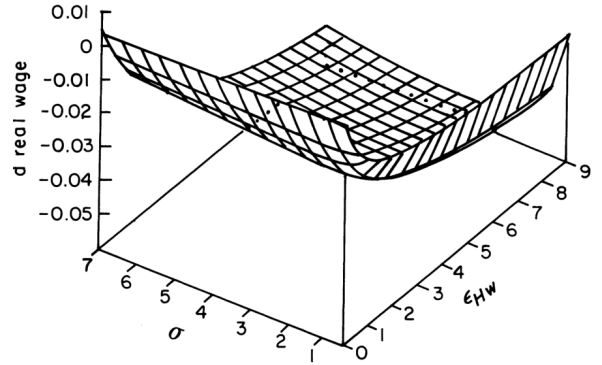


FIG. 7.—Sensitivity of the first-period real-wage response to σ and ϵ_{Hw}

the higher ϵ_{Hw} is. But for firms to actually increase their labor input,

Figure 7: Real-wage sensitivity to σ and ϵ_{Hw} .

7.6 Figures 8 and 9: Responses to the orthogonal military-composition shock

Read:

- Figure 8 shows the response of private value added to the shock denoted η^p .
- Figure 9 shows the response of private hours to the same shock.
- This shock changes the composition of military spending rather than overall purchases.
- The models do not match these responses as well as the main military-purchase shock.

Detailed interpretation: The composition shock is less central to the paper's argument because it may be measured with more error and may be associated with labor-supply or patriotism effects. The empirical response shows private output and hours increasing even though the models predict weaker or negative responses.

Economic message: The main evidence for the model comes from purchase shocks, not from military-composition shocks.

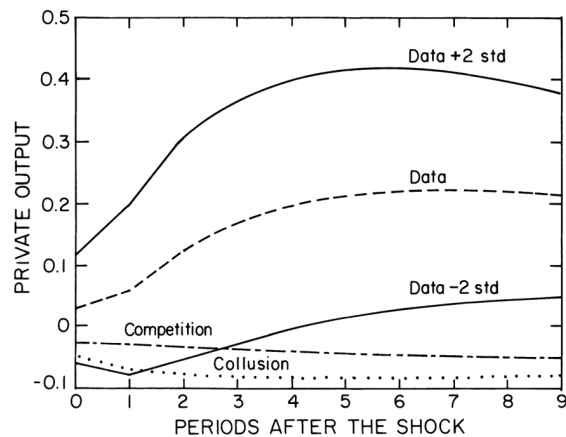


FIG. 8.—Response of private value added to η^p

Figure 8: Private value added to η^p .

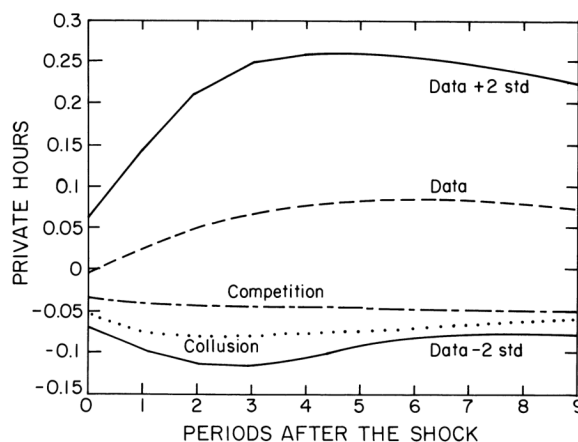


FIG. 9.—Response of private hours to η^p

Figure 9: Private hours to η^p .

7.7 Figures 10 and 11: Real wages to composition shocks and long historical robustness

Read:

- Figure 10 shows real wages falling in the data after the military-composition shock, unlike the main purchase shock.
- Figure 11 plots annual changes in defense purchases, real output, and real wages from 1891 to 1970.
- The long historical plot shows large defense-purchase movements around wars.
- Output and real wages tend to rise around major defense expansions, supporting the postwar evidence at a broad level.

Detailed interpretation: Figure 10 reinforces the caution about the composition shock. Figure 11 provides a lower-frequency robustness check: the broad pattern that defense purchases and real wages rise together is visible in a much longer sample, even if the prewar data are less clean and subject to price controls and measurement problems.

Economic message: The main postwar result is not an isolated sample artifact, but historical evidence is supportive rather than decisive.

FIG. 9.—Response of private hours to η_t^p

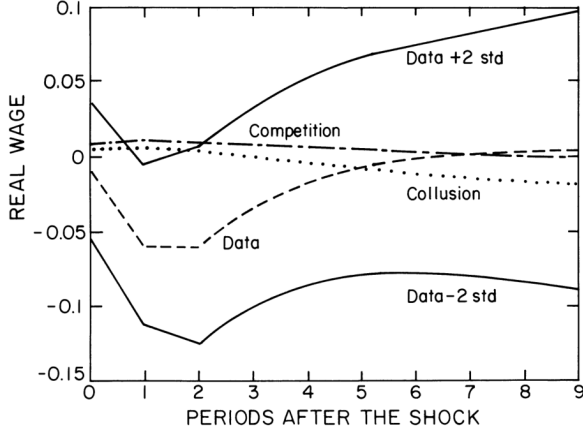


Figure 10: Real wages to η^p .

of real wages from 1891 to 1988. The real GNP series comes from

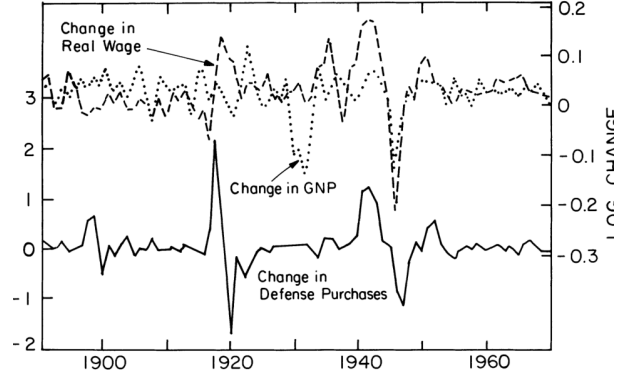


FIG. 11.—Changes in defense purchases, output, and real wages, 1891–1970

Figure 11: Defense purchases, output, and real wages, 1891–1970.

8 Typed theoretical result blocks

8.1 Result block 1: Competitive demand shocks do not shift labor demand

In the competitive benchmark,

$$F_H(K_t, H_t, z_t) = w_t.$$

Because K_t and z_t are fixed with respect to the demand shock, aggregate demand does not move labor demand. Employment can rise only if labor supply shifts.

8.2 Result block 2: Markup variation shifts labor demand

With imperfect competition,

$$F_H(K_t, H_t, z_t) = \mu_t w_t.$$

A decline in μ_t raises desired labor input at any real wage, shifting labor demand outward.

8.3 Result block 3: Collusive markups fall in booms

Current demand booms increase the gain from undercutting rivals. To satisfy incentive compatibility in the repeated oligopoly game, firms must choose a lower markup in high-demand states.

8.4 Result block 4: The real wage test

A labor-supply explanation predicts falling real wages after a demand shock. A markup-based labor-demand explanation predicts rising real wages. The empirical military-spending response favors the markup-based mechanism.

9 Detailed result map

- **Empirical fact:** military purchase shocks raise private output and real wages.

- **Competitive failure:** competitive models can generate output responses, but not the positive real-wage response.
- **Oligopoly mechanism:** high demand makes collusion harder to sustain and lowers the incentive-compatible markup.
- **Labor-market implication:** a lower markup shifts labor demand outward, raising hours and real wages.
- **Sensitivity evidence:** output and wage responses increase with the markup and markup cyclical.
- **Preference limitation:** stronger labor-supply responses help output but imply falling real wages.
- **Robustness:** long historical data broadly show defense spending, output, and real wages rising together.

10 Short summary

- Rotemberg and Woodford study how aggregate demand affects real activity in a dynamic general-equilibrium model with oligopolistic pricing.
- The model embeds repeated collusion within industries in an otherwise RBC-like framework.
- Booms make collusion harder to sustain because firms have more to gain from undercutting prices.
- The optimal collusive agreement lowers markups in booms.
- Lower markups shift labor demand outward and can raise output, hours, and real wages.
- Military purchase shocks are used as relatively exogenous aggregate demand shocks.
- The competitive model has trouble matching the positive real-wage response.
- The oligopolistic model matches the real-wage response much better.
- The paper provides a real-strategic alternative to pure nominal-rigidity explanations of demand-driven fluctuations.

11 Conclusion

11.1 Core conclusion

The paper shows that imperfect competition and endogenous markups can explain why aggregate demand shocks have real effects. Increases in military purchases raise private output and real wages, a pattern that is difficult for the competitive benchmark but natural when markups fall in booms.

11.2 Economic intuition

In a collusive oligopoly, firms keep prices high only because future punishment deters deviation. When current demand is high, the temptation to cut price and capture sales is stronger. To keep collusion incentive-compatible, firms lower current markups. Lower markups increase labor demand and can raise both employment and real wages.

11.3 Incremental contribution revisited

The paper's incremental contribution is to make variable markups endogenous inside a dynamic general-equilibrium model and then discipline the mechanism with calibrated impulse responses to military-spending shocks. It bridges repeated-game industrial organization and aggregate business-cycle analysis.

11.4 Implications

The results suggest that demand shocks can matter even without nominal rigidities if market power and strategic pricing are present. They also imply that markup cyclicalities are a central object for understanding business cycles.

11.5 Limitations

The model is calibrated rather than fully structurally estimated. Military spending is relatively exogenous but not perfectly clean. The military-employment composition shock behaves anomalously. The historical robustness exercise is supportive but less precise than the postwar exercise.

11.6 Takeaway

The paper's lasting message is that market structure changes macroeconomic transmission. Aggregate demand can shift labor demand when imperfectly competitive firms reduce markups in booms.